DR. ANIKA SHARMA

Scientist E — Atmospheric Instrumentation & Applied AI

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PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

TECHNICAL DOMAINS

- Avionics (MIL-STD-1553 / ARINC 429, Avionics integration)
- Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems)
- Embedded systems (Embedded C, ESP32 / STM32 microcontrollers)
- Edge AI (NVIDIA Jetson deployment)
- Drones & UAV systems (UAV payload integration)
- AI / machine learning (PyTorch, Deep learning)
- IoT (MQTT and Node-RED, InfluxDB and Grafana)
- Computer vision (Image super-resolution, Object detection)
- GPU computing (CUDA / GPU computing)
- Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation)
- Software & tools (Python, Docker and Linux)
- Technical leadership (Team leadership)

PROFESSIONAL EXPERIENCE

Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory — New Delhi

Scientist E — Jul 2016 – Present

Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

Tools: Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop — Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** — Jan 2021 – Present

Avionics Integration Group, Halcyon Avionics Design Bureau — Bengaluru

Senior Design Engineer — Jun 2004 – Jun 2016

Integration and qualification testing of airborne electronics and UAV payloads.

Tools: MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.

Test Systems Group, Meridian Electronics Limited — Bengaluru

Graduate Engineer — Aug 2002 – May 2004

Built automated test equipment for avionics line-replaceable units.

Tools: LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

TECHNICAL PROJECTS

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Edge super-resolution pipeline for UAV imagery (Ongoing) — Sep 2021 – Present

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Fictional sample profile — not a real person

EDUCATION

Northfield Institute of Science and Technology
Ph.D., Electrical and Electronics Engineering — 2021 – ongoing
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

Coastal Institute of Aeronautical Engineering — Chennai
M.Tech., Avionics — 2010
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

Northfield Institute of Science and Technology, Northfield University — Pune
B.E., Electronics and Communication Engineering — 2004
Grade: First Class with Distinction

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, “Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs,” Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, “An IoT telemetry framework for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, “Edge telemetry dashboards for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, “A low-cost GNSS receiver chain for radiosonde tracking,” Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

LEADERSHIP & RESPONSIBILITIES

Kestrel Atmospheric Instrumentation Laboratory
Supervisor, M.Tech. interns in applied machine learning — 2022 – Present
Supervise summer and dissertation interns on edge-AI projects.

Kestrel Atmospheric Instrumentation Laboratory
Chair, Technical Evaluation Committee for radiosonde procurement — Apr 2023 – Feb 2024
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

CERTIFICATIONS & TRAINING

Example Academy of Applied Computing
Deep Learning for Edge Devices — Mar 2022
Duration: 40 hours
Credential: EAAC-DL-0042
Valid until: Mar 2029

Example Conference on Instrumentation (ECI 2023)
Workshop on Efficient Deep Learning for Embedded Vision — Dec 2023
Duration: 2 days

AWARDS & HONOURS

Example Conference on Instrumentation (ECI 2023)
Best Paper Award — 2023

TECHNICAL SKILLS

Avionics	MIL-STD-1553 / ARINC 429, Avionics integration
Scientific instrumentation	Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
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Edge AI	NVIDIA Jetson deployment
Drones & UAV systems	UAV payload integration
AI / machine learning	PyTorch, Deep learning
IoT	MQTT and Node-RED, InfluxDB and Grafana

LANGUAGES

English, Hindi

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